

UEI514: IoT BASED SYSTEMS

L	T	P	Cr
2	0	2	3.0

Course Objectives: The objective of the course is to provide basic understanding of Internet of Things, its architecture, Sensor modules, communication protocols, platforms and its deployment, applications, latest trends and one case study.

Introduction: Introduction to Internet of Things, characteristics and benefits of IoT, basic architecture, OSI reference model, IoT gateways and IoT cloud facilities and applications of IoT.

IoT Sensors modules and applications : Introduction to IoT based measurements, Smart sensors, MEMS based sensors, Pyroelectric PIR motion detector, Ultrasonic range finder, Gas sensor, 3-axis gyro sensor module. Digital transducers, Encoders, Touchpad. Finger print scanner. Hall-Effect, IR range finder, light dependent resistor LDR module,

IoT Actuators and Gateways : Solid state relay, dc motor, servo motor and stepper motors. Arduino and Raspberry Pi as IoT Gateways.

Simplified IoT Architecture : The core IoT Functional Stack, Layer 1: Things: Sensors and Actuators Layer, Layer 2: Communications Network Layer, Layer 3: Applications and Analytics Layer.

IoT Protocols and Clouds: Networking of sensors, Need of communication protocols for IoT systems, introduction and comparison of protocols like HTTP, CoAP, MQTT, AMQP, IFTTT, Li-Fi and BLE etc. IoT cloud, its need and characteristics. Introduction and comparison of various clouds like AWS, Azure, Blynk, Watson, Google cloud and Thinger.

Laboratory Work:

1. Interface various sensor modules like LDR.
2. Exhibit the knowledge Ultrasonic, Hall effect, PIR, Gyro, digital encoder etc. with Arduino board.
3. Deployment of an embedded system on IoT cloud using USB serial and /or Wi-Fi modules.

Course Learning Outcomes (CLO):

Student will be able to:

1. Exhibit the concept and benefits of IoT
2. Exhibit the knowledge of sensor modules for IoT applications
3. Elucidate IoT actuator, gateways and architectures.
4. Implement IoT system using protocols and IoT cloud

Text Books:

1. *IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things*, Robert Barton, Patrick Grossetete, David Hanes, Jerome Henry, Gonzalo Salgueiro by Cisco Press, 2017.
2. *Internet of Things: A Hands-on Approach*, by Arshdeep Bahga and Vijay Madisetti Universities Press, 2014.

Reference Books:

1. *Internet of Things, Abhishek S Nagarajan, RMD Sundaram Shriram K Vasudevan, Wiley India 2019.*
2. *Internet of Things, Raj Kamal, McGraw Hill Education 2017*

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST+EST	70
2	Sessional (May include Assignments//Quizzes/Lab Evaluations)	30